Skypedia

Founded in 2026

Crew Members:

-Haochen Huang

-Mark Lin

-Paolo Dal Cin

Core System Features

Country Search

Dynamic retrieval of global stats including population, currency, and official identifiers.

Live Weather

Integration with meteorological APIs to provide capital city temperatures in real-time.

Smart Costing

Automated pricing engine based on geographic distance and duration of stay.

API Used

Skypedia works by using three free web tools (APIs) that talk to each other:

- **REST Countries API**: This tool provides all the facts about a country, such as its official name, population, languages, and map coordinates.

  https://restcountries.com/v3.1/name/{name}

- **Open-Meteo API**: We send the coordinates from the first tool to this weather API to get the real-time temperature of your destination.

  https://api.open-meteo.com/v1/forecast?latitude={lat}&longitude={lon}&current_weather=true

- **Flag CDN**: This service automatically displays the correct national flag based on the country you search for.

  https://flagcdn.com/w320/{code}.png

Real-Time Forecasts

The Open-Meteo Integration

Weather is the most critical variable in travel planning. Skypedia automatically retrieves current temperatures for destination capitals during the search phase.

This allows our travel agents to provide immediate packing advice and seasonal recommendations directly from the dashboard.

The Calculation Engine

$0.12

PER KILOMETER TRANSIT

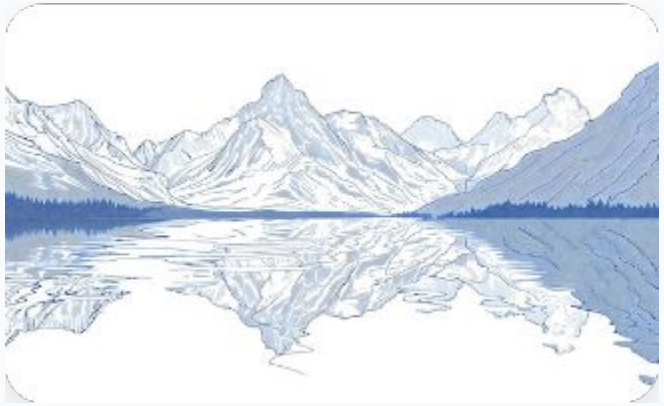
Logic Behind the Costs

Our pricing model is transparent and scalable. We utilize the Haversine approximation to calculate travel distances between coordinates.

$$\text{Cost} = \text{dist} \times 0.12 + \text{days} \times 150 + 200$$

dist= distance
days= days for staying
200= stable price

Supported Global Regions



Europe

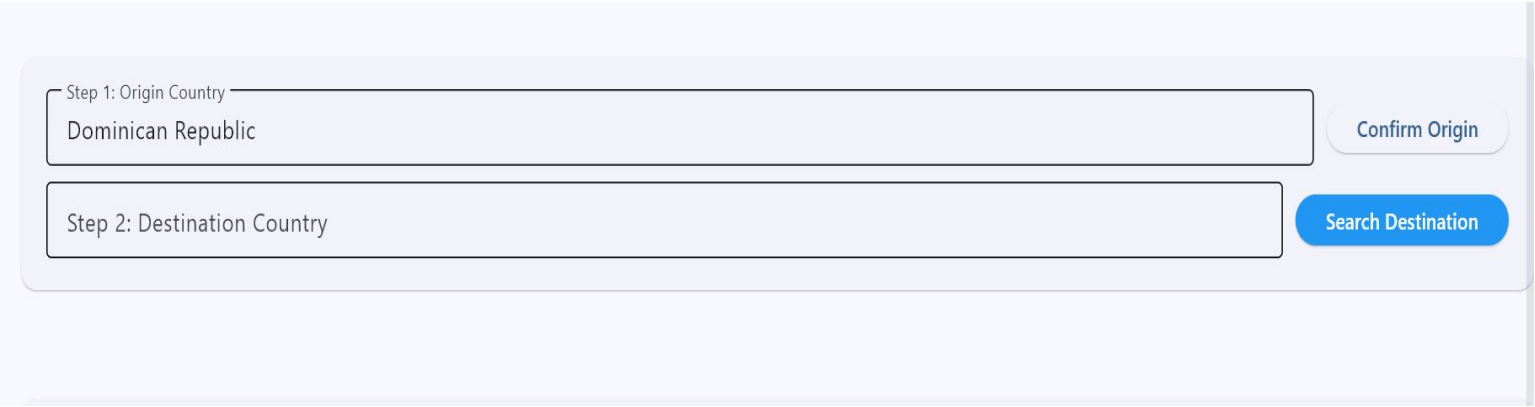
Landmarks

Tropical

Client Booking Records

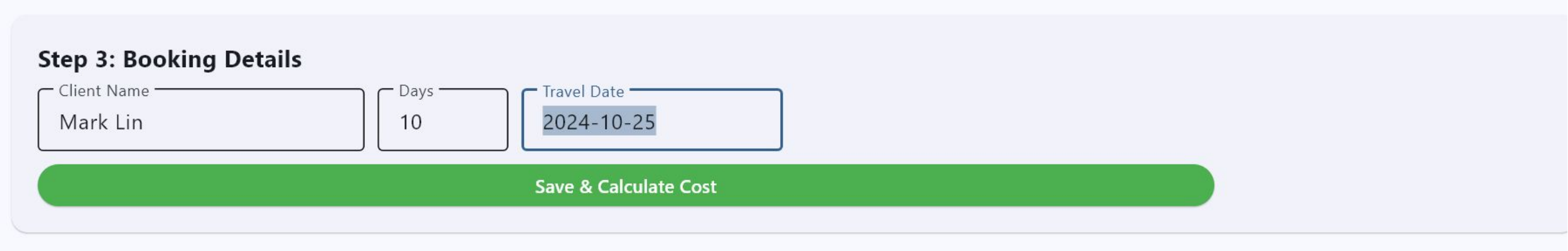
Our JSON-based data persistence layer ensures client records remain accessible between sessions.

Client Name	Origin	Destination	Travel Date	Total Cost
Mark Lin	Dominican Republic	China	2024-10-25	$4049.57
Haochen	Nepal	FInland	2024-10-12	$2609.28
Paolo	Australia	New Zealand	2024-12-01	$2277.08

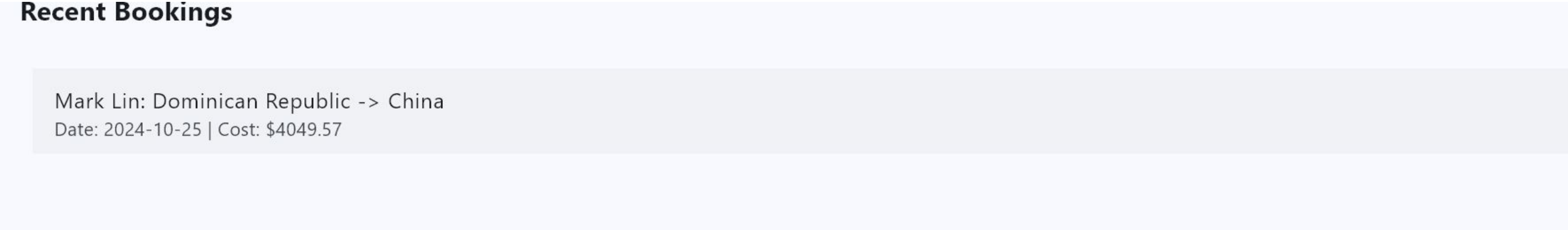


Step 1: User writes their country origin.

Step 2: User writes their destination. After pressing “Search Destination”, Information about the country, such as the Weather, Population, Capital, etc…



Step 3: User writes their information, such as their names, how many days are they staying, and the date.



Step 4: After pressing “Save & Calculate Cost”, from step 3, it will show you the information you have already inputted, and its costs for the ride.

Application User Journey

1

Set Origin

Establish home base location coordinates.

2

Global Search

Query destination APIs for real-time stats.

3

Calculate

Process distance and duration pricing.

4

Finalize

Save itinerary to system database.